Extending Sentiment Analysis with Advanced Transformer Models: A Comparative Study

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***Abstract*—The interpretation of user sentiment together with public opinion depends on sentiment analysis technology as an essential component of modern digital media platforms such as social media and financial news as well as entertainment reviews. This research exam- ines how transformer-based models namely BERT and its advanced versions RoBERTa and DistilBERT achieve better classification results when integrated with deep learning layers such as CNN, RNN, and BiLSTM. This study aims at determining which model architecture shows optimal results for negative sentiment along with neutral sentiment and positive sentiment instead of developing one universal solution. A normalized prepro- cessing method was used to prepare multiple datasets while pretrained transformer tools segmented the texts into tokens. The hybrid model architectures received contextual embeddings as their input. Evaluation used accuracy and precision together with recall and F1-score as well as ROC-AUC to measure performance. A real-world application of the results enables optimized sentiment detection by picking appropriate models for de- tecting precise emotional sentiments which benefits brand monitoring as well as financial analysis and customer experience systems**

***Index Terms*—BERT, RoBERTa, NLP, DistilBERT,Sentiment Analysis**

I. Introduction and Statement of the Problem

In today's digital world, everyone is going to express their ideas and concepts through websites and network platforms. Sentiment analysis helps computers to clarify text-based human emotions along with a system that evaluates written statements for positive or negative expressions. So, this analysis method classifies verbalization into three emotional forms: neutral, negative, or positive. Numerous businesses depend on this data tool to track brand reputation along with customer feedback for better service development. The huge amount of wide data sets becomes possible through artificial intelligence and machine learning, so sentiment analysis systems make it better for organizations with good choices and public sentiment understanding. Sentiment analysis analyzes emotions that occur in text data through an automated method which results in positive and negative and neutral classifications. The tool functions as a leading Natural Language Processing (NLP) tool which businesses primarily use to acquire knowledge about public mood and customer satisfaction alongside web-related trends. 1 Historically, sentiment analysis mostly depends on the machine learning models, but due to the changing of technology and trends performing with transfer models, like BERT, got the most rapidly improved performance. But also, there is an increasing delight in searching for other transformer models like RoBERTa, and DistilBERT. In this project, we will take sentiment analysis further by testing different transformer models with var- ious datasets to see which one works best. This research is important because it covers essential steps in data science, like gathering data, cleaning it up, building models, and evaluating their performance.It allows hands-on experience with deep learning frameworks like TensorFlow/PyTorch and NLP models. It enhances understanding of transformer ar- chitectures and sentiment analysis techniques, relevant for various industry applications.

The core NLP application called sentiment analysis allows organizations to decode client opinions along with public attitudes. Traditional machine learning models like support vector machines and logistic regression struggle with contextual understanding, espe- cially in social media. Those methods often did not have the ability to interpret the actual meaning of words, especially when the context was modified according to how the sentence was structured. The accuracy of sentiment analysis using Transformer models like BERT, RoBerta, and DistilBERT can change depending on the dataset and the spe- cific task. In this research, the team combines these models with other techniques like CNN,RNN, and BiLSTM to improve how well they predict sentiment. The primary diffi-

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culty, thus, is finding out which combination of model-deep learning method—including CNNs, RNNs, or BiLSTMs—is best in a specific context. Our project counters this difficulty by experientially attempting model combinations with data from various do- mains like finance, movies, and social media to determine the most effective and efficient configuration for sentiment prediction.

A. Key challenges addressed includes

This paper analyzes the performance capabilities of BERT, RoBERTa, and Distil- BERT through different dataset measurements. Deep Learning Integration: Assessing CNN, RNN, and BiLSTM's impact on performance. The evaluation found the ideal model for all social media and finance and movie review platforms. The strategy of find- ing an optimal balance between precise results and machine system resource management for practical deployment purposes

II. Review of Literature

The study by Bello et al. (2021) demonstrated that combining BERT with CNN, RNN, and BiLSTM improves accuracy, precision, recall, and F1-score in sentiment analysis tasks. The study by Jayakody et al. (2021) analyzed customer reviews on Twitter to determine whether people had positive, negative, or neutral opinions. They utilized computer models, such as Support Vector Machine (SVM) for grouping similar items, and logistic regression for identifying patterns in data. Researchers found that the logistic regression model performed best when paired with a technique called CountVectorizer, which measures word frequency. In simple words, the researchers found that a particular method was able to predict setiments of a tweet correctly with accuracy of 88.262 The researchers Chirrini et al. (2021) created two models based on BERT for classifying text from twitter ans similar platforms. Their models were able to get 92accuracy, showing that BERT works really well for finding out the sentiment of posts. They pointed out BERT's ability to understand the context of words in both directions mode it better at predicting sentiment than older deep learning methods.By fine tunning BERT with data specific to the topic, they were able to make it even more accuracy for analyzing sentiments of social media. Another study by Habbat et al. (2021) and his team handled the problem of analyzing sentiment in datasets where some type of sentiments are much more common than others. They proposed a method combining BERT with various deep learning models to improve accuracy. Their approach effectively balanced the results, enhancing BERT's ability to handle datasets with uneven sentiment distributions.

III. Objectives of the Study

The main objectives of this study are:

A. Primary Objectives

1) To build a model that uses BERT, RoBERTa, and DistilBERT with CNN, RNN, and BiLSTM to improve sentiment classification.
2) To compare how well different models like the the combination of BERT, RoBERTa, DistilBERT with CNN, RNN, and BiLSTM perform in identifying positive, nega- tive, and neutral sentiments.
3) o measure the accuracy, precision, recall, and F1 score of different sentiment anal- ysis models to see which one works best.
4) To visualize the results using confusion matrices, and ROC curves.

IV. Data Collection

As a part of Data collection, we gathered different public datasets from Kaggle dealing with social media (Twitter sentiment), financial news headlines, and movie reviews. The datasets had different structural formats where text and sentiment data used various columns—review and text for the first dataset, while the second dataset had news as well as label and sentiment as column names for corresponding fields. We executed a column renaming operation on every dataset that used tweet for text data and sentiment for label data to achieve database integration. The data concatenation process became possible through pandas because we standardized all datasets into a single master DataFrame structure while preserving data information.

We conducted a second step after renaming to eliminate all rows that included null values to keep the data stable. We converted sentiment values into numbers where Nega- tive was set to 0, Neutral was set to 1, and Positive received the value of 2. So all datasets maintained uniform labeling. The sentiment categories needed proportional representa- tion during sampling to avoid class imbalance from affecting model training. The combination of cleaned datasets produced a combined dataset with about 1,126,062 record instances. We first cut down the dataset to 100,000 rows through the stratified sampling technique to maintain sentiment stability because of BERT and RoBERTa's 3 resource-intensive training requirements. The revised dataset ended in reduced data.csv and established the base for our hybrid sentiment classification models with transformers connected to CNN, RNN, and BiLSTM structures. The preprocessing method estab- lished standardized procedures for defining a dataset that enabled fair assessment of models across various textual areas. **Dataset Sources:**

- Twitter Entity Sentiment Analysis
- Sentiment Analysis for Financial News
- Twitter Sentiment Dataset (3 million labelled rows)
- Sentiment Analysis Dataset

V. Architecture

Sentiment analysis architecture with BERT, RoBERTa, and DistilBERT requires different components for its operation. The initial text goes into pre-trained transformer models including BERT, RoBERTa, or DistilBERT. The models extract semantic word and sen- tence meaning through processing their input text to create embedded representations. After passing the embeddings to a downstream classifier (like the

fully connected neural network or other models) the system produces sentiment predictions. Each transformer model offers distinct benefits.

BERT (Bidirectional Encoder Representations from Transformers): It offers two-directional text processing which enables it to extract word meanings from their contextual context.

RoBERTa (Robustly optimized BERT approach): Presents itself as an en- hanced BERT version which optimizes training procedures and enhances performance across multiple NLP applications.

DistilBERT: This compact version of BERT delivers a combination of speed and efficiency alongside strong accuracy throughout its operations particularly in situations with constrained processing power.

A. Training and Evaluation

The models will be evaluated based on metrics such as accuracy, precision, recall, and F1 score. A confusion matrix will serve to display and analyze classification errors so the identification of incorrectly categorized sentiments can be performed.

B. Implementation Framework

Python and deep learning frameworks like TensorFlow and PyTorch will be used in this research. Other libraries such as Hugging Face's Transformers, NLTK, Pandas, and Scikit-learn will be used in preprocessing the text. All model training time and computing efficiency will be optimized using GPU acceleration.

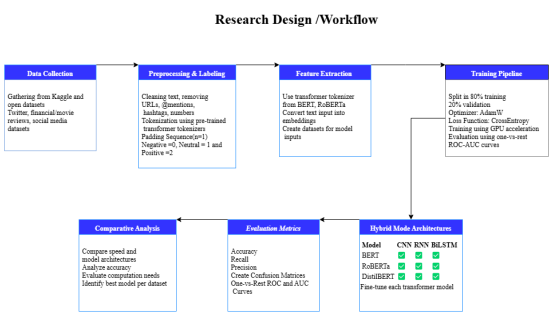

Fig. 1. Work flow of Sentiment analysis models

VI. Exploratory Data Analysis (EDA) and Hypotheses

The goal of Exploratory Data Analysis (EDA) in this project is to take a close look at the tweet data before building any models. It helps us understand what the data looks like, how the tweets are spread out across different emotions, and if there are any problems like missing or messy text. By spotting patterns and cleaning the data, we make sure it's ready for the machine learning model to learn from it and make better predictions.

A. Label Encoding and Tweet Length Calculation

To prepare the textual sentiment labels for machine learning models, we performed label encoding and calculated tweet lengths as follows:

- **Label Encoding**:The original dataset contains sentiment labels in textual format: *Negative*, *Neutral*, and *Positive*. Since machine learning models require numerical input, we mapped these labels to numeric values using a dictionary:

```
sentiment_map = {"Negative": 0,
"Neutral": 1, "Positive": 2}
df['label'] =
df['sentiment'].map(sentiment_map)
```

  This created a new column label in the dataframe with encoded values: 0 for Negative, 1 for Neutral, and 2 for Positive.
- **Tweet Length Calculation:** o analyze the length of each tweet (after cleaning), we computed the number of words in each tweet using:

```
df['tweet_length'] = df['clean_tweet']
.apply(lambda x: len(x.split()))
```

  This generated a new column , which stores the word count of each tweet. This information is useful for text preprocessing steps such as padding or truncating sequences for input into deep learning models.

B. Sentiment Distribution Visualization

A simple bar chart was created to show how many tweets fall under each sentiment category: **Negative**, **Neutral**, or **Positive**. First, the chart was styled to look neat and clear using a white grid background and defined figure size. The data set was then scanned to count the number of tweets that belonged to each sentiment class. These counts were displayed as bars on the chart. Instead of showing numerical labels like 0, 1, and 2, the x-axis was labeled with the actual sentiment names to make it more user-friendly. The chart also included a title and axis labels for better understanding. Finally, the chart was displayed, helping us easily identify which sentiment appears most often and whether the data is evenly distributed across the three categories.

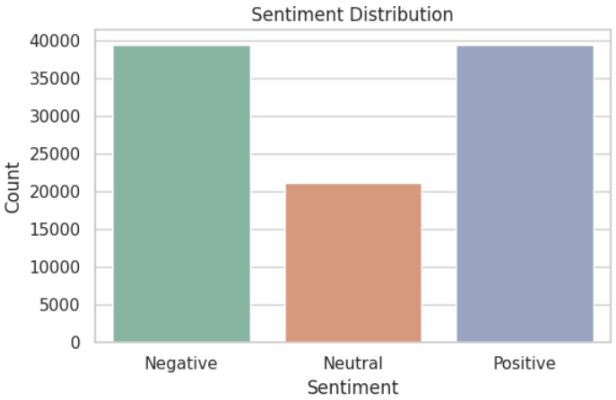

Fig. 2. This chart shows that people mostly tweet either positive or negative things, and fewer tweets are neutral

C. Tweet Length Distribution

This code creates a simple chart to show how many words people usually use in their tweets. It helps us see if most tweets are short, long, or medium in length by counting how often each tweet length appears.

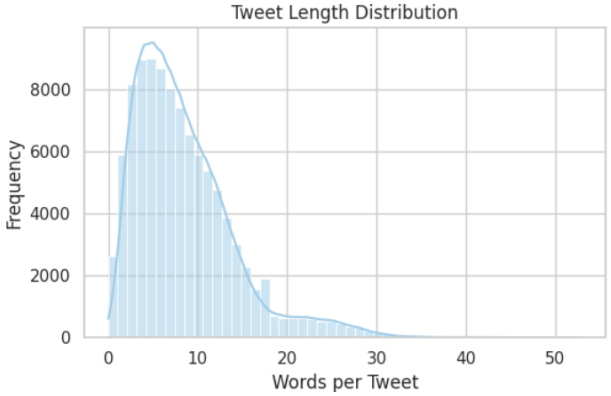

Fig. 3. This chart shows how many words people usually use in their tweets. It helps us see if most tweets are short, long, or medium in length.

D. Word Clouds by Sentiment

Word clouds were used to show which words appear most often in Positive, Neutral, and Negative tweets. A word cloud is a visual representation of text where the size of each word indicates how frequently it appears — the more often a word is used, the larger it looks in the image. In the code, a function was first defined to create and display a word cloud. Then, the dataset was divided into three groups based on sentiment: Positive, Neutral, and Negative. All the tweets in each group were combined into one large text string, and a separate word cloud was generated for each. These word clouds provide a quick and easy way to understand the most commonly used words in different types of tweets just by looking at the pictures.

Fig. 4. This picture shows the most used words in positive tweets—words like "love," "thank," and "today" appear bigger because people use them more often.

Fig. 5. This picture shows the most used words in negative tweets—bigger words like "miss," "work," and "go" appear more often in the text.

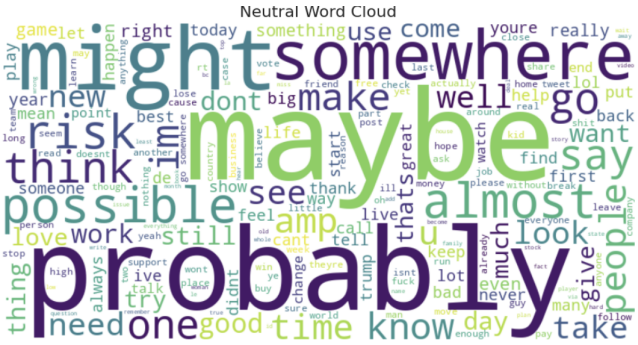

Fig. 6. This image shows the words people use most in neutral tweets—words like "probably," "maybe," and "might" are bigger because they're used more often.

E. Top Common Words by Sentiment

The top 20 most used words in Positive, Neutral, and Negative tweets using bar charts. It first gathers all the words from each group of tweets and counts how many times each word appears. Then, it creates a chart where longer bars represent words that appear more frequently. These charts help us quickly understand which words are most commonly used in each type of sentiment.

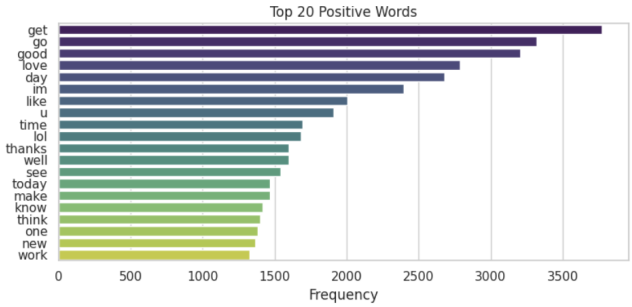

Fig. 7. This chart shows the top 20 most used words in positive tweets, with "get," "go," and "good" being the most common.

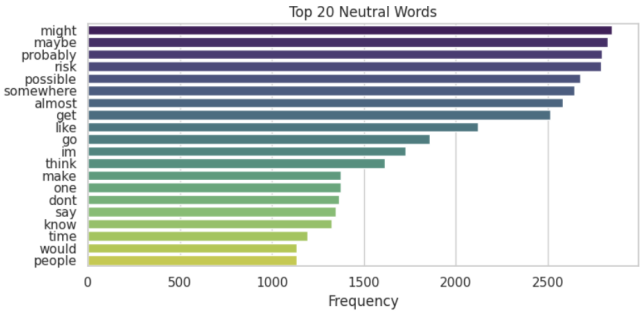

Fig. 8. This image shows the words people use most in neutral tweets—words like "probably," "maybe," and "might" are bigger because they're used more often.

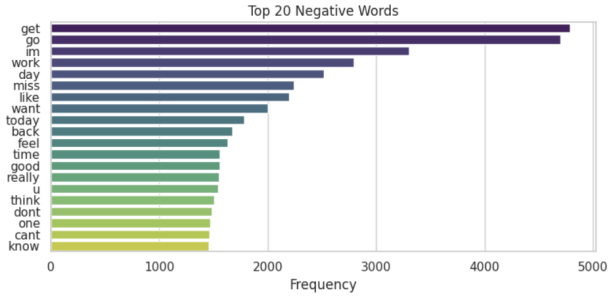

Fig. 9. This chart shows the words people use most in negative tweets—words like "get," "go," and "I'm" are used the most often.

F. BERT Tokenization and Input Preparation

The code gets tweets ready so BERT can understand them. Since BERT can't read normal text, the tweets are turned into numbers using a tool called a tokenizer. It also 9 adds special symbols, makes all tweets the same length, and creates a guide to help BERT focus only on the actual words and ignore the extra fillers. In the end, it gives three things: the number version of tweets, a guide to what's important, and the sentiment labels — all set for BERT to start learning.

VII. Data Analytics

As a precursor task of our sentiment analysis work, we conducted an extensive data analytics task to get to know our dataset's behavior and structure. The data set consists of 100,000 tweets where each sentiment was labeled as positive, neutral, or negative. Following an initial exploratory data analysis (EDA) process, the data was approximately balanced with a distribution of approximately 39.4and 21.2training and fair performance evaluation for all models. We also learned various tweet properties such as text length, token distribution, and content density that allowed us to define input parameterizations for transformer models such as maximum sequence length and padding scheme.

Our preprocessing pipeline processed the entire text by converting it to lowercase while removing URLs along with emoticons, user mentions, and special characters, and executing stopword elimination and lemmatization. Our cleaning operations produced top-quality token embeddings for all pre-trained models, including BERT, RoBERTa, and DistilBERT. Restructuring the analysis stage enabled the identification of patterns that demonstrate when particular sentiments emerged alongside particular language styles. The structured information contained within unstructured text served as a foundation for creating reliable future predictions through data analysis processes.

The data analysis approach served as much more than an initial preparation process within our methodology. Data analytics served as both a step for managing text data and ensuring classes were balanced while optimizing transformer inputs and maintaining consistent high-quality data between model comparison tests. Our understanding of the data set strengthened our approach to model development since we knew that the distribution was balanced and the text was appropriate for effective sentiment analysis. This framework formed the basis for deriving accurate performance conclusions regarding which nine hybrid deep learning models would best handle multiple sentiment categorization activities.

VIII. Results and Visualization

In this section, we evaluate and visualize the performance of nine sentiment classification models models developed using BERT, RoBERTa, and DistilBERT combined with CNN, RNN, and BiLSTM architectures. The evaluation metrics used are:

1) **Accuracy:** Proportion of correctly predicted labels over total predictions.
2) **Precision:** Measures how many of the predicted sentiments are correctly identified.
3) **Recall:** Measures how well the model identifies all true positive examples.
4) **F1-Score:** Harmonic mean of precision and recall, indicating balance between them.
   The following table summarizes the classification performance per sentiment class for all nine models

TABLE I
MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision			Recall			F1-Score		
		NEG	NEU	POS	NEG	NEU	POS	NEG	NEU	POS
BERT-CNN	0.791	0.78	0.91	0.74	0.72	0.93	0.79	0.75	0.92	0.76
BERT-RNN	0.723	0.68	0.85	0.70	0.72	0.80	0.68	0.70	0.82	0.69
BERT-BiLSTM	0.713	0.67	0.83	0.70	0.71	0.77	0.69	0.69	0.80	0.69
RoBERTa-CNN	0.794	0.74	0.92	0.78	0.80	0.92	0.73	0.77	0.92	0.75
RoBERTa-RNN	0.695	0.69	0.77	0.67	0.67	0.75	0.69	0.68	0.76	0.68
RoBERTa-BiLSTM	0.699	0.69	0.75	0.67	0.65	0.81	0.69	0.67	0.78	0.68
DistilBERT-CNN	0.784	0.73	0.93	0.76	0.78	0.91	0.72	0.76	0.92	0.74
DistilBERT-RNN	0.724	0.69	0.85	0.69	0.69	0.82	0.70	0.69	0.84	0.70
DistilBERT-BiLSTM	0.740	0.72	0.80	0.74	0.74	0.84	0.70	0.73	0.82	0.72

As seen in the table, BERT-CNN and RoBERTa-CNN achieved the highest overall accuracy and strong performance across sentiment classes. DistilBERT models were ef- ficient and relatively balanced, with DistilBERT-BiLSTM performing well in identifying both negative and positive sentiments. RNN-based models generally showed moderate performance, while BiLSTM improved the recall slightly, particularly for positive sen- timents. The following confusion matrices visu- ally represent model predictions across sentiment categories:

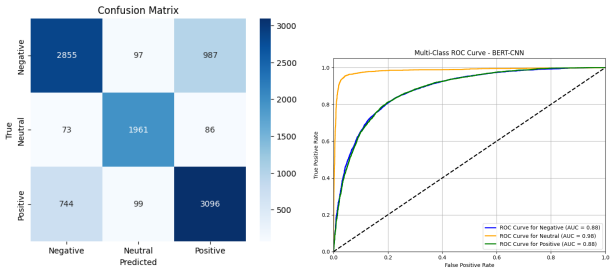

Fig. 10. Confusion Matrix - BERT-CNN: Out of 4939 negative tweets, 2855 were correctly classified. Out of 2120 neutral tweets, 1961 were correctly classified. Out of 3939 positive tweets, 3096 were correctly classified. The AUC scores (Negative = 0.88, Neutral = 0.98, Positive = 0.88) indicate strong class separation, especially for neutral sentiment.

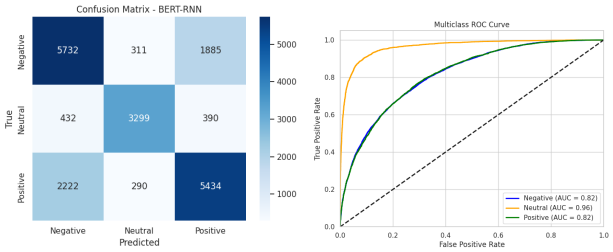

Fig. 11. Confusion Matrix - BERT-RNN: Out of 7928 negative tweets, 5732 were correctly classified. Out of 4121 neutral tweets, 3299 were correctly classified. Out of 7946 positive tweets, 5434 were correctly classified. With AUCs of 0.82 (Negative), 0.96 (Neutral), and 0.82 (Positive), BERT-RNN moderately distinguishes sentiments, performing best on neutral.

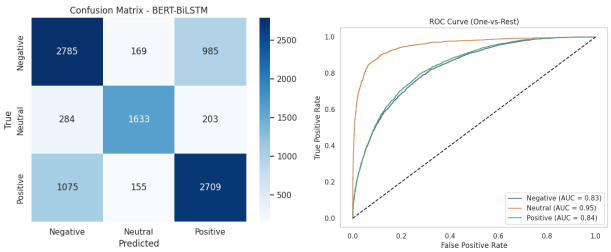

Fig. 12. Confusion Matrix - BERT-BiLSTM: Out of 3939 negative tweets, 2785 were correctly classified. Out of 2120 neutral tweets, 1633 were correctly classified. Out of 3939 positive tweets, 2709 were correctly classified. AUCs of 0.83 (Negative), 0.95 (Neutral), and 0.84 (Positive) show improved balance and better recall over RNN for all sentiments.

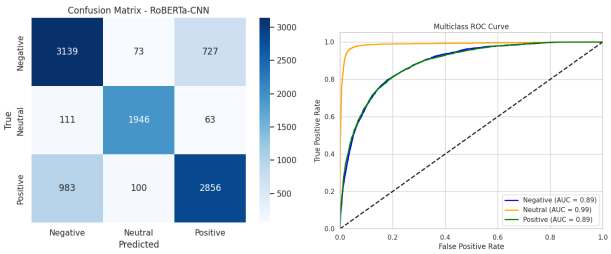

Fig. 13. Confusion Matrix - RoBERTa-CNN: Out of 3939 negative tweets, 3139 were correctly classified. Out of 2120 neutral tweets, 1946 were correctly classified. Out of 3939 positive tweets, 2856 were correctly classified. High AUCs—0.89 (Negative), 0.99 (Neutral), and 0.89 (Positive)—demonstrate excellent class distinction and top-tier performance.

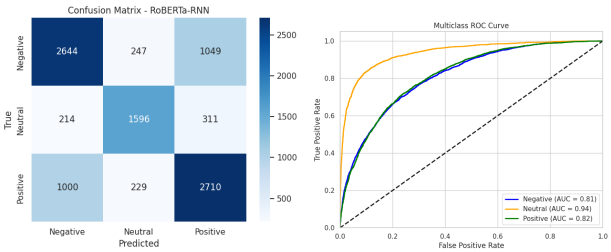

Fig. 14. Confusion Matrix - RoBERTa-RNN: Out of 3939 negative tweets, 2644 were correctly classified. Out of 2120 neutral tweets, 1596 were correctly classified. Out of 3939 positive tweets, 2710 were correctly classified. With AUC values of 0.81 (Negative), 0.94 (Neutral), and 0.82 (Positive), this model struggles with class separation compared to CNN.

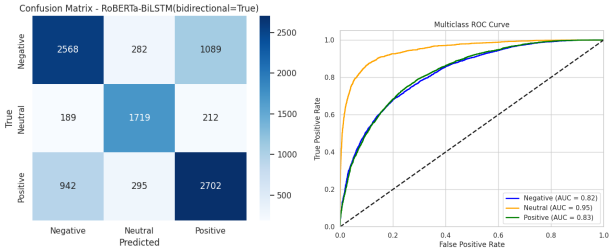

Fig. 15. Confusion Matrix - RoBERTa-BiLSTM: Out of 3939 negative tweets, 2568 were correctly classified. Out of 2120 neutral tweets, 1719 were correctly classified. Out of 3939 positive tweets, 2702 were correctly classified. AUCs of 0.82 (Negative), 0.95 (Neutral), and 0.83 (Positive) reflect balanced performance, especially for neutral detection.

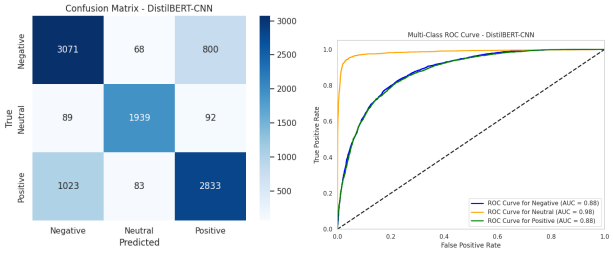

Fig. 16. Confusion Matrix - DistilBERT-CNN: Out of 3939 negative tweets, 3071 were correctly classified. Out of 2120 neutral tweets, 1939 were correctly classified. Out of 3939 positive tweets, 2833 were correctly classified. AUC values of 0.88 (Negative), 0.98 (Neutral), and 0.88 (Positive) indicate high performance with efficient sentiment discrimination.

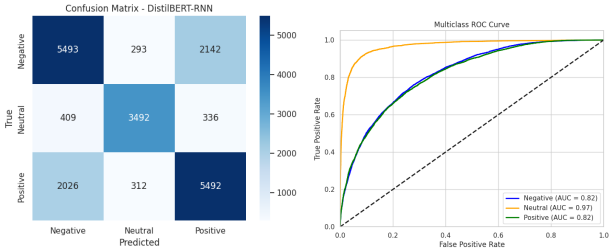

Fig. 17. Confusion Matrix - DistilBERT-RNN: Out of 7928 negative tweets, 5493 were correctly classified. Out of 4121 neutral tweets, 3492 were correctly classified. Out of 7928 positive tweets, 5492 were correctly classified. DistilBERT-RNN yields AUCs of 0.82 (Negative), 0.97 (Neutral), and 0.82 (Positive)

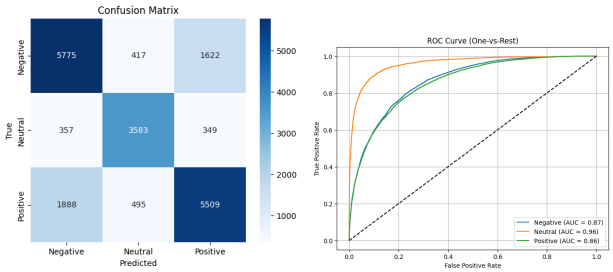

Fig. 18. Confusion Matrix - DistilBERT-BiLSTM: Out of 7928 negative tweets, 5775 were correctly classified. Out of 4121 neutral tweets, 3583 were correctly classified. Out of 7928 positive tweets, 5509 were correctly classified. AUC scores—0.87 (Negative), 0.96 (Neutral), 0.86 (Positive)

IX. Conclusion

This study aimed at comparing nine transformer-based sentiment classification models which include BERT, RoBERTa and DistilBERT used together with CNN, RNN, and BiLSTM. We tested each of the models based on combination of different evaluation metrics such as accuracy, precision, recall, F1-score and ROC-AUC coefficient using a variety of data sets such as tweets, financial news and movie reviews. From the results, it is evident and clear that RoBERTa-CNN has relatively performed well among all other models. BERT-CNN had higher accuracy in neutral sentiment detection, and DistilBERT-BiLSTM model was good at both positive and negative sentiment detection having strong balance between efficiency and accuracy. These findings indicated that there is no model that is said to be ideal for all types of sentiment. However, it is important to remember that the selection of the model depends on the type of sentiment target and availability of resources. Adding transformer embeddings with CNN, RNN, BiLSTMs enhance the accuracy of sentiment classification and hence using a hybrid model is useful for practical purposes.

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